

Dakshita Wadibhasme

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Education

IIT (ISM) Dhanbad

Bachelor of Technology in Computer Science (GPA: 8.88 / 10.00)

Expected May 2028

Dhanbad, Jharkhand

- **Relevant Coursework:** Data Structures and Algorithms (C), Unix and software tools, Computer programming, Algorithm design and analysis, Operating System, DBMS, Computer Organisation and architecture, Statistical methods

Bhavan's B.P. Vidya Mandir (Class X) 99.20% (C.B.S.E. Board)

May 2022

Dr. Ambedkar College (Class XII) 94.33% (Maharashtra State Board)

May 2024

Projects

Nimbus | React, Node.js, MongoDB Atlas, RAG, Gemini API, JWT

[Github](#)

- Engineered a Retrieval-Augmented Generation (RAG) pipeline utilizing MongoDB Vector Search and Gemini Vision to embed and retrieve past design assets, ensuring aesthetic consistency in AI-generated media.
- Developed a full-stack orchestrator with a React frontend and Node.js backend to automate administrative workflows, including AI-driven Markdown report synthesis and integrated SMTP for automated email dispatch.
- Designed a secure, scalable architecture featuring JWT multi-session authentication with refresh token rotation, alongside cloud-based media hosting via Cloudinary.

Systems Engineering & Concurrency | C/C++, xv6 OS, Socket API, POSIX Threads

[Server GitHub](#) | [xv6 GitHub](#)

- **Custom HTTP Server:** Engineered a high-performance, multi-threaded web server in C++ using native Socket APIs to efficiently manage concurrent client connections and process TCP data streams.
- **Kernel Multithreading:** Implemented kernel-level multithreading in the xv6 operating system by designing custom `clone` and `join` system calls in C.
- **Concurrency Optimization:** Bridged user and kernel spaces by developing a user-level threading library, demonstrating deep comprehension of thread lifecycles, network protocols, and shared memory.

Technical Skills

Languages: C, C++, Python, JavaScript, HTML/CSS

Frameworks & Databases: React.js, Node.js, Express.js, Bootstrap, MongoDB, RAG pipelines

Tools & Systems: Linux, POSIX Threads, Git, Docker, REST APIs, JWT, VS Code

Achievements

- Qualified JEE Advanced with Rank 3118.
- Scored 99.69 percentile in MHT-CET and got 2nd position in the RTMNU university Ladies category merit list.
- Bharti Airtel Merit-cum-Means Scholarship recipient for the full duration of the B.Tech program.
- Specialist in Codeforces(Max rating: 1453).
- Won the best girls team prize at Obscurathon 1.0 (relay-style hackathon conducted by CSE department)
- Finalist in the National Entrepreneurship Olympiad at high school level conducted by E-cell Visvesvaraya National Institute of Technology (VNIT), Nagpur.
- Among top 3 finalists in British Council Project at International level (Documentary making)

Social Engagements

Member: CSE society of IIT Dhanbad, WebCSE(Supervisor), CSE Times (Creative head)

Volunteer: Member of Vidharbha Youth Organization's Nature Club,Anchor for UDBHAV 2025 (the departmental annual program)

Sports-Engagements: Badminton-Represented college at **Inter IIT sports** and also at Infinito sports event at IIT Patna, 1st Winner in College Badminton Tournament